

AroMR: Decentralizing Olfactory Displays into the Environment for Olfactory-Augmented Experiences in Mixed Reality

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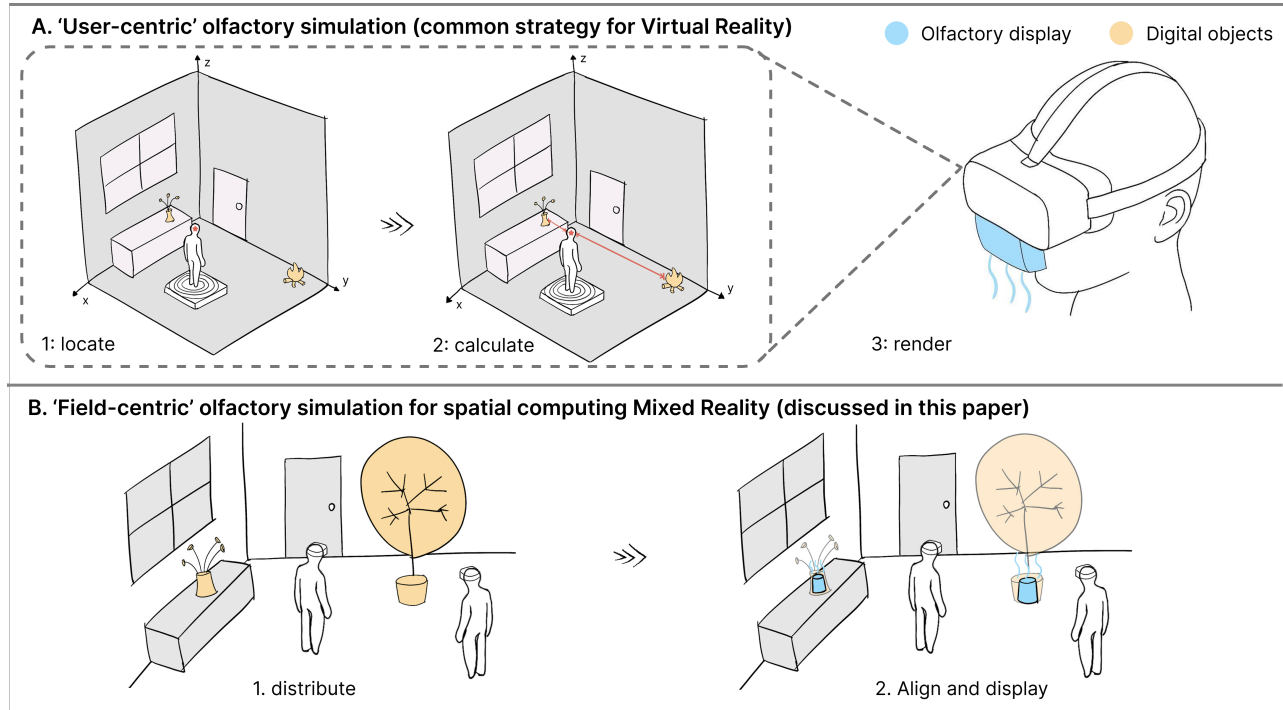


Figure 1: We propose an idea that decentralizes olfactory displays from the user's body into the environment to provide olfactory feedback in mixed reality experiences. (A). Sketch map of the current "user-centric" olfactory simulation strategy, including locate, calculate, and render; (B). "field-centric" olfactory simulation strategy proposed in this paper for mixed reality

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CHI EA '25, Yokohama, Japan
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ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3720290>

Abstract

Smell, as a vital sensory modality, has the potential to significantly enhance user experience in immersive environments; however, its application in mixed reality (MR) remains underexplored. Current olfactory displays (ODs) are typically fixed to head-mounted displays (HMDs) as active olfactory inputs, limiting interaction and

failing to capture the dynamic nature of human olfactory perception. To better integrate olfactory experiences into MR, we propose a "field-centric" strategy that decentralizes ODs into the physical environment as passive outputs. We further introduce AroMR, a proof-of-concept system that employs this strategy to incorporate olfactory feedback into MR experiences. A user study involving eight participants interacting with AroMR across three pre-designed scenarios revealed the potential of our approach to enhance engagement and create more immersive MR experiences. Based on these findings, we propose a conceptual design space to guide future research on integrating olfactory elements into MR.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **User interface toolkits**.

Keywords

Mixed Reality, Olfactory Interface, Interaction Techniques

ACM Reference Format:

Yibo Wang, Ziqi Liu, Jiao Xue, and Qi Lu. 2025. AroMR: Decentralizing Olfactory Displays into the Environment for Olfactory-Augmented Experiences in Mixed Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720290>

1 Introduction

Nowadays, Mixed Reality (MR) technologies have shown great promise in creating visually immersive hybrid environments, and with devices like the Apple Vision Pro¹ and Meta Quest 3², users can now easily access a future where digital objects seamlessly blend with the physical environment [9]. However, smell as an interaction modality with the potential of enhancing realism in immersive experiences [5], deepening emotional connections [4], and providing unique contextual information [29], has been underexplored in MR experiences. Current olfactory interactions are not yet suitable for adaptation to MR experiences, as they have largely remained limited to pure Virtual Reality (VR) scenarios [12, 14, 19, 24], where the physical environment plays a minimal role. We hereby pose the question: How to effectively incorporate olfactory experiences into MR experiences?

Current olfactory displays (ODs) for immersive experiences are typically affixed to head-mounted displays (HMDs) [17] or designed as lightweight devices attached to the body [6, 19]. These studies represent a similar olfactory simulation strategy that we conclude as "user-centric" (as shown in Figure 1 A). Such designs often offer limited interaction, thus failing to simulate the dynamic and spatial nature of human olfactory perception, such as wafting the scents [34]. What if, instead of a "user-centric" strategy, we could adopt a "field-centric" approach by decentralizing ODs from the user's body to the entire environment (as shown in Figure 1 B), thus transforming the olfactory interface from a wearable device to an environmental system? By aligning digital objects with ODs, we could endow them with natural olfactory properties, as odors dispersed in the physical environment could naturally interact with

airflow and body movements. The idea of leveraging physical environment in digital olfactory experience has been actively explored in the literature [30], however, much of the previous research has not examined the potential relationship between this approach and immersive environments, leaving open questions about how this strategy can be adapted to MR.

To quicker understand how such ideas could scale and how difficult the development process could be, this paper contributes a proof-of-concept system named AroMR (a blend of Aroma and MR) that enables users to interactively experience olfactory-augmented MR based on the "field-centric" strategy. The system consists of multiple customized ODs that function as olfactory outputs when distributed in the environment, and a corresponding MR software platform that runs on Meta Quest 3. In our study, we invited eight participants to engage with three pre-designed scenarios and conducted semi-structured interviews to explore their experiences. The preliminary qualitative feedback indicated that participants were largely positive about the testing experience and noted that the steps to configure the ODs are considered to be easy to follow. Although some participants reported that imprecision in MR spatial scanning negatively impacted their experience, overall, our system effectively provided proper olfactory feedback, contributing to more engaging MR interactions. Based on further iterative explorations, we also learned that the "field-centric" strategy can go beyond the simple scenarios tested in the user study and there are a lot of untapped opportunities for this idea. For example, we found that ODs can provide haptic feedback, creating natural and novel olfactory experiences that correspond to interactions through physical contacts [24]. Moreover, distributing ODs throughout the ambient environment allows for the natural utilization of airflow generated from physical movements. To further explore this potential, we discuss possible design showcases and exemplary scenarios, which we hope will inspire further investigation within the research community.

2 Related Work

2.1 Multi-sensory Interactions in Mixed Reality

As a new emerging field, MR [27] is revolutionizing interaction paradigms. The incorporation of technologies such as LiDAR and depth cameras into HMDs has enabled spatially aware capabilities, allowing the collection and usage of spatial data to foster immersive, natural, and intelligent multi-modal interactions [32]. However, current HCI research focusing on MR has predominantly concentrated on visual and auditory stimuli [18], overlooking the significant potential of olfactory inputs to enhance user experience in such experiences. MR disperses virtual content broadly across the physical space, allowing for more natural interactions with the environment [7, 8]. As a result, users in MR scenarios are typically more attuned to the interactions between virtual content and the physical environment, such as how virtual objects are anchored to real-world surfaces or how physical factors (like airflow and daylight) influence the virtual content. The physical environment has long been interference for fully immersive VR, while this gap between physical environment and VR is naturally addressed in MR, as physical environment plays as a foundational layer of MR [29]. This key difference has inspired a strategy discussed in this paper

¹<https://www.apple.com/apple-vision-pro/>

²<https://www.meta.com/quest/quest-3/>

on designing olfactory experiences for MR by actively leveraging the physical environment [9, 10].

2.2 Olfactory Display for Immersive Experiences

The HCI community has actively explored olfactory interfaces for fully immersive environment [3, 19, 22–24]. With a preliminary literature review, it can be concluded that current olfactory simulation methods mainly adopt a certain strategy we hereby call "user-centric", as it typically focuses on a "point" as the olfactory render objective, mainly the user's nose. This strategy often involves first locating the odor receptor as a "point" in a three-dimensional space, then determining the olfactory input for the receptor through calculations (e.g. computational fluid dynamics [15, 21]), and finally using digital olfactory display interfaces near the nose to display the odor (as shown in Figure 1 A.). OD technologies used in the systems, such as ultrasonic atomization [1, 2], or heating techniques utilized in systems like ScenThread [31], or flexible olfactory displays [19], each faces unique challenges in scalability, miniaturization, and user experience customization.

However, these applications are mostly fixed to VR HMDs [12, 17, 20, 21, 25, 26, 33] or users' body [19], which often leads to limited interaction capabilities and fails to capture the dynamic and spatial nature of human olfactory perception [16, 34]. For example, if designing a VR chemical experiment simulation, actions such as waving a hand over a container to direct odors toward the nose cannot be accurately rendered, as most fixed olfactory displays cannot simulate air currents and subtle changes in odor concentration that correspond with gestures. We presume that this limitation has been largely overlooked due to VR's tendency to present content in a concentrated, high-density format, leading users to disregard nuanced interactions, such as the effects of airflow [13]. As a result, there is a need for alternatives to this "user-centric" approach for better immersive experiences in MR. This study contributes a "field-centric" strategy to bridge this gap by shifting the olfactory interface to a room-scale system to naturally preserve the nuanced interactions.

3 AroMR: Proof-of-Concept System

This study proposes a proof-of-concept system named AroMR to explore how the "field-centric" strategy can function in MR scenarios. The AroMR system consists of two parts: an OD hardware, and a corresponding MR platform with three pre-designed scenarios. When using the system, the user first selects the virtual object via the interface (see figure 2.B1) and place it in the physical environment using intuitive hand interactions (see figure 2.B2 & B3). Then, the user needs to place the OD on the physical location corresponding to the virtual object (see figure 2.B4). The Meta Quest 3 will then communicate with the OD via WiFi, and the OD at the corresponding location will be triggered once the user toggles the status on the interface, thus emitting the scent corresponding to the digital object to endow it with olfactory properties. This process could be repeated to distribute digital objects and ODs accordingly to create an olfactory-augment MR environment.

3.1 Technical Implementation

We have designed an OD that can remotely communicate with Meta Quest 3 via a Wi-Fi connection. The OD has a diameter of 8 cm and a height of 15 cm, and all parts are fabricated from 3D-printed PLA plastic, except for the piezo holder, which is made of PTFE for more effective cleaning. As shown in Figure 2A, the device includes an odor display module with a piezoelectric atomizer, an odor storage module featuring a scent reservoir and cotton stick for liquid odor absorption, and a PCB module. These components are housed within an inner shell contained in an outer shell with two layers: the upper layer holds the scent storage and playback components, while the lower layer contains the PCB, which includes the control board, relays, atomizer driver boards, and a battery. Prior to use, olfactory liquid (e.g., essence or perfume) saturates the cotton stick before contacting the atomizer. When powered, the atomizer disperses the odor into the air by creating a thin film of the ingredient. An ESP32 WROVER Module micro-controller automates the device using the Arduino IDE and features a USB-C interface for charging and code uploading, while operating offline with a lightweight 3.7V lithium battery.

3.2 MR Scenarios with Olfactory Experience

To further validate the implementation of the "field-centric" strategy in MR scenarios, we developed three representative MR scenarios with different features: "Potted Plant", "Flowering Tree", and "Window View" (as illustrated in Figure 3). The scenarios were developed on Unity and run on the Meta Quest 3, utilizing Meta's room recognition functionality for physical plane identification and localization. In the "Potted Plant" scenario, a small virtual potted plant is situated on an indoor desk, with an OD placed in the same location to simulate its scent. This scenario aims to test the fundamental interactive logic of the "field-centric" strategy. The "Flowering Tree" scenario features two ODs positioned on the upper and lower shelves of an indoor bookshelf, emitting the scents of flowers and grasses to replicate the aroma of a virtual flowering tree. This setup is designed to investigate how the interplay of different odor sources influences participants' perceptions. Finally, in the "Window View" scenario, the outdoor landscape is transformed into a virtual forest scene, with an OD positioned at the open window to utilize natural airflow for simulating the forest scent. This scenario aims to explore how the combination of ambient airflow and the virtual forest scent affects participants' perception.

4 User Study

We tested with eight participants (male: 6, female: 2; none non-binary or self-described; $M = 25.1$ years; $SD = 3.1$) with either olfactory design expertise or MR design experience recruited from our laboratory. The participants used the Meta Quest 3 and the ODs to experience the three scenarios under verbal instructions. The experimenter had completed room scanning and calibration in advance. After putting on the headset, participants sequentially selected three scenarios in the floating interface to experience. In the first two scenarios, users could freely place virtual objects and ODs on any suitable physical surface. After participants completed their experiences with all three scenarios, the experimenter conducted a semi-structured interview to gather insights on their overall user

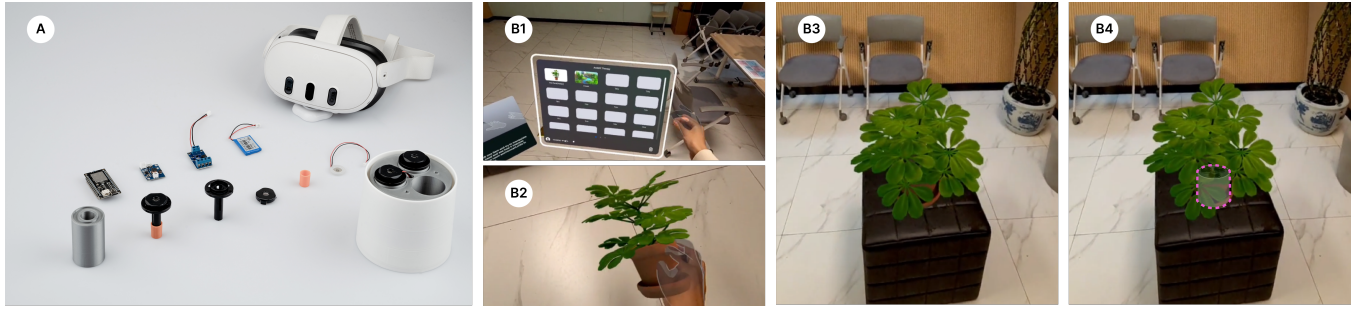


Figure 2: (A) Assembly of the AroMR system. (B) The usage process of the system: (B1) The user selects the digital object via the MR interface. (B2) The user uses hand control to place the digital object. (B3) The user aligns the digital object with the OD (hidden under the digital object and not visible). (B4) The processed version of B3 to demonstrate the relationship between the digital object and the OD (highlighted in pink).

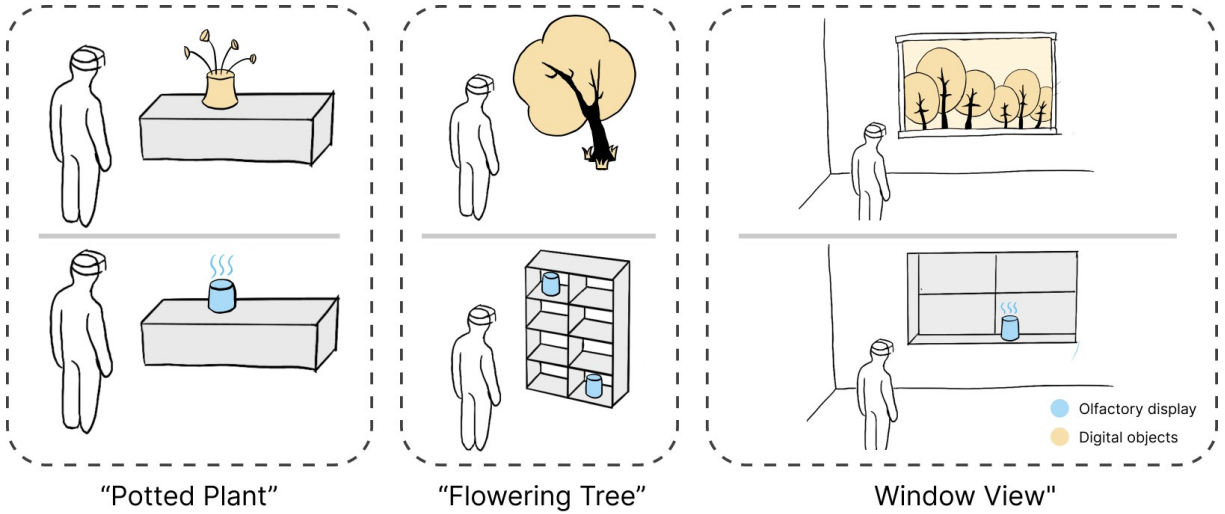


Figure 3: Three typical scenarios designed for the user study.

experience. Each session lasted for around an hour. The interview was guided by the following questions:

To assess whether the digital objects were effectively aligned with the ODs in the physical environment, we asked: **Q1:** Do you perceive that the odors emanate from the virtual objects? **Q2:** Do you think the process of aligning ODs with virtual objects took a long time?

To investigate users' perceptions regarding the interplay of different odor sources, we posed: **Q3:** Are you able to associate the odor source with a specific physical location? **Q4:** If yes, does perceiving the direction of the odor source enhance the realism of the virtual objects?

Finally, to explore the potential benefits of integrating natural airflow into olfactory interactions, we asked: **Q5:** Do you think it is appropriate to introduce airflow factors into olfactory interactions within MR scenarios? **Q6:** Do you believe that incorporating airflow factors into olfactory interactions adds more design possibilities?

All interviews were recorded, later transcribed, and translated into English. They were then analyzed line by line by the first two

authors. Drawing from the qualitative feedback gathered through interviews and observation of the experiment process, we identified several key insights and design recommendations regarding the "field-centric" strategy.

5 Results

Here we provide summaries about the findings of our user study from three perspectives: the feasibility of aligning ODs with digital objects, perception of the spatiality of odor, and odor's interaction with natural airflow.

5.1 Feasibility of Aligning ODs with Digital Objects

In the "Potted Plant" scenario, users were supposed to align the virtual plant with the ODs in space by putting both of them in the same position. Participants reported that "virtual plant can be placed by hands in MR just like real objects, which makes the alignment process much simpler and more intuitive" (P3), and did not require "additional calibration processes or supporting structures"

(P7). Additionally, in the "Window View" scenario, where the virtual environment in MR serves as a substitution for the real-world space, participants were able to directly align the ODs with the correct position based on the real-world references (e.g. windowsill). Participants described this process as being "basically natural and easy to follow" (P5). However, in the post-study discussion, P7 remarked that "the current OD is too large to be seamlessly integrated and concealed behind various virtual objects". In summary, despite the flaws, MR has preliminarily enabled the merging of virtual objects with the physical environment, which facilitates a straightforward process of aligning virtual objects with ODs, making the proposed interactive system of the "field-centric" strategy fundamentally feasible.

5.2 Perception of the Spatiality of Odor

In the "Flowering Tree" scenario, participants reported that they first experienced the interplay of the two different odor sources, then located each odor source by further sniffing. This process enhanced their "sense of the tree's height and size" by smelling "flower from the top of the tree, while grass and soil from the bottom" (P2), thus allowing for a "sense of embodiment in the MR experience" (P1). In the "Window View" scenario, the participants' experiences were particularly pronounced. They reported that the perception of odor direction significantly enhanced their understanding of "the spatial relationship between the virtual and real scenes" (P3). Through this directional perception, they were able to "sense that the odor was coming from outside the window, rather than being a mere stimulus that only varied in intensity" (P6). This indicates that the spatiality of smell can influence spatial perception and the overall immersive quality of MR environments. The perception of the spatiality of odor can serve as a cue for users to sense external space, reconstructing their perception and understanding of space, thereby enhancing the realism of MR experiences.

5.3 Odor's Interaction with Natural Airflow

Participants largely agreed that incorporating natural airflow into the olfactory experience in MR scenarios will be novel and challenging. Physical world characteristics such as airflow "constitute an external interference for scenes like VR, but can become a part of the experience in MR scenarios where the virtual and the real are blended" (P4). Participants believed that the varying states of airflow in different physical spaces could enhance the authenticity of their olfactory experiences, stating that "even within a closed indoor room, the process of odor generation and diffusion can increase its sense of realism" (P3). Furthermore, the tactile sensations brought by airflow can complement the olfactory experience. In the "Window View" scenario, participants noted that "the cool breeze by the window allows me to feel that this is a smell coming from the forest" (P2). In addition, some participants believed that the introduction of airflow could enhance the possibilities at the design level and the fun of interaction in olfactory experiences. Users could "feel the disturbance of odors by their hands and even interact with odors through actions such as fanning with a fan" (P1).

6 Discussion and Design Space

Based on the testing results, we further reflect on the design opportunities provided by the "field-centric" strategy to effectively incorporating olfactory experiences into more MR experiences. We organized the ideations and illustrated them as exemplary MR scenarios for our discussions (as shown in Figure 4).

6.1 Reconstructing Perception of Daily Spaces in MR

When leveraging the "field-centric" strategy, users can place ODs in familiar daily spaces, then reconstruct their perception and understanding of the original physical space in MR with olfaction as a cue. While previous research equipped furniture with swarm robots to reconfigure the physical environment to match the virtual content, our idea is to make use of people's perceptions of daily spaces without reconfiguring any [28]. For example, as shown in Figure 4, users can build small enclosed spaces with items such as boxes, and reconstruct their cognition of the space inside the box through olfactory stimulation upon opening it. In this manner, everyday activities such as opening a box or pushing and pulling cabinet doors can be reinterpreted with new meanings in MR. Similarly, users can leverage physical environments (e.g. windows and walls) to merge the visual stimuli of MR with the tactile feedback from surfaces like window sills, along with olfactory cues. This combination has the potential to transform users' original perception of the physical space and expand it into a hybrid external realm.

6.2 Interaction through Physical Contacts

As in the "field-centric" strategy, ODs are decentralized into the environment instead of fixed to the body, which allows ODs to serve as passive haptic props [11], providing haptic feedback and allowing for more intuitive olfactory interaction. Current researches have focused on making ODs more flexible (e.g. small, attachable) while maintaining its robustness [19]. This advancement would allow for more different placement strategies based on the specific objects being simulated. As a result, in addition to fixed physical surfaces, ODs could be attached to movable objects or being directly manipulated by the user. For example, movable items like a glass of wine can be represented either by affixing small ODs to the cup or by employing graspable and portable ODs. This would allow odor to be naturally "aired out" by the hand movements when moving the OD to the face, thus creating responsive and intuitive interactions. This was also discussed in previous research as "enactive smelling" [24].

6.3 Utilizing Natural Airflow

Previous study has investigated using artificial airflow and thermal stimulator along with olfactory feedback to provide multi-modal experiences [26]. At the same time, natural airflow as an equally important aspect has long been treated as interference [29]. With "field-centric" strategy, instead of considering it as an interference, users can create a diverse range of olfactory experiences and interactions by utilizing natural airflow generated from various types of physical movements in MR. For instance, as shown in Figure 4, airflow generated by appliances like air conditioners and fans, or natural sources such as windows and doors, can be combined with

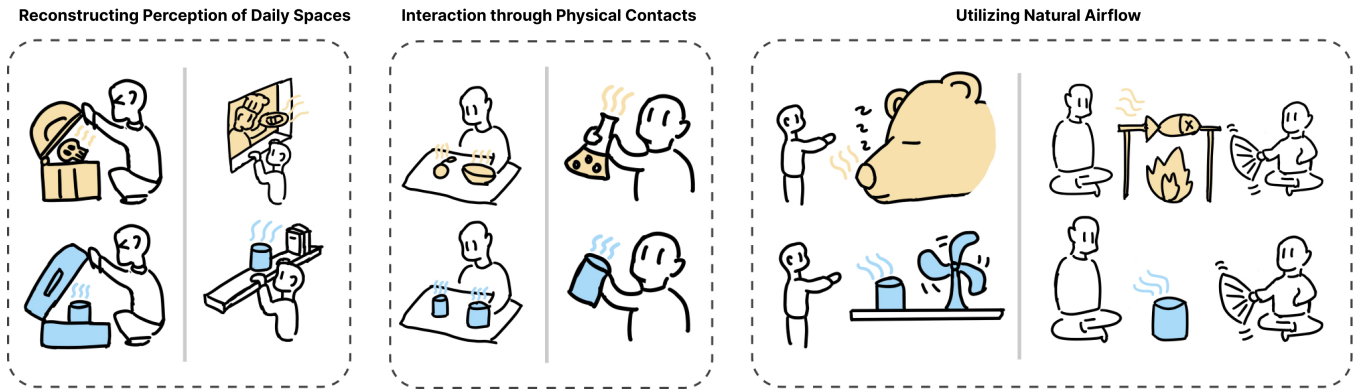


Figure 4: The design space of "field-centric" strategy. We illustrated two scenarios for each topic to show the rich design opportunity.

odor emissions to manipulate the direction of scents, resulting in distinct olfactory experiences. Furthermore, varying intensities of airflow and different ventilation areas can be utilized to simulate different virtual objects and settings. Users can also actively interact with odors through actions such as disturbing or fanning, allowing them to recreate specific smells, like that of a campfire.

7 Limitation and Future Work

Although we discussed several implications of our work as described in the above paragraphs, we also need to acknowledge some limitations of this study.

Firstly, the limitation in the ODs. In our expectation, the ODs should be able to be placed on physical surfaces more than horizontal planes (e.g. walls, door), as well as attached to movable objects (e.g. cups). However, current OD design is neither compact enough nor specifically designed to be attached to the places mentioned above. Future iterations of the study should center on the improvement of the hardware system design.

Secondly, current alignment of digital objects with physical ODs requires purely manual adjustments, which leads to complicated manual calibrations and unavoidable inaccuracies that finally hurt the overall user experience. In future work, we aim to develop features for the automatic alignment of digital objects with ODs by utilizing advancements in MR technologies.

Finally, our user study design did not incorporate a comparison between the proposed "field-centric" strategy and the traditional "user-centric" strategy, primarily due to time constraints. Future iterations of the study should prioritize including such a comparative analysis.

8 Conclusion

We present a "field-centric" olfactory simulation strategy, and applied it in designing AroMR, a proof-of-concept system, where ODs are decentralized into the environment to provide natural, on-demand olfactory feedback for more immersive MR experiences. We conducted a user study based on our system, from which we

collected qualitative feedback regarding the system, discussed potential design recommendations, and concluded design space for future designers and researchers who wish to use this strategy.

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